Deep Learning

Introduction

Deep Learning is a subset of Machine Learning and Artificial Intelligence that uses neural networks with multiple layers to learn from data. It is inspired by the human brain and helps computers perform intelligent tasks such as image recognition, speech recognition, language translation, and prediction.

What is Deep Learning?

Deep Learning is a technique where computers learn automatically from data using Artificial Neural Networks. It is called deep because it contains multiple hidden layers between the input and output layers.

Neural Network Architecture

Input Layer → Hidden Layer(s) → Output Layer

Working of Deep Learning

- Data Collection
- Data Preprocessing
- Model Training
- Prediction
- Evaluation

Types of Deep Learning Networks

Artificial Neural Network (ANN)

Artificial Neural Network (ANN) is the basic type of Deep Learning network inspired by the human brain. It consists of neurons connected through layers such as input layer, hidden layer, and output layer. ANN is mainly used for prediction and classification problems.

Applications:
- Student Result Prediction
- Spam Detection
- Sales Prediction

Convolutional Neural Network (CNN)

Convolutional Neural Network (CNN) is mainly used for image and video processing tasks. It automatically detects features such as edges, shapes, and objects from images.

Applications:

- Face Recognition
- Object Detection
- Medical Image Analysis

Recurrent Neural Network (RNN)

Recurrent Neural Network (RNN) is designed for sequential data where previous information is important. RNN remembers previous inputs using memory.

Applications:
- Speech Recognition
- Language Translation
- Text Prediction

Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is an advanced version of RNN that can remember information for a longer time. It solves the memory limitation problem of RNN.

Applications:
- Weather Forecasting
- Time Series Prediction
- Stock Market Prediction

Activation Functions

Activation Functions decide whether a neuron should activate or not. Common activation functions are Sigmoid, ReLU, and Tanh.

Advantages of Deep Learning

- High accuracy
- Handles large datasets
- Automatic feature extraction
- Learns complex patterns

Disadvantages of Deep Learning

- Requires large datasets
- Needs powerful hardware
- Training takes more time
- Difficult to understand

Applications of Deep Learning

- Healthcare
- Banking
- Education
- Transportation
- Entertainment

- Robotics

Conclusion

Deep Learning is one of the most advanced technologies in Artificial Intelligence. It helps machines learn automatically from data and solve real-world problems efficiently.